

Future Collider Perspectives on Higgs CP Violation

ML-learned observable in ZH and ttH at future linear Higgs Factory

Christoph Englert(U of Manchester), Jay Nesbitt(U of Glasgow),

Andrew Pilkington(U of Manchester), Aidan Robson(U of Glasgow),

[Yuyang Zhang\(U of Glasgow/DESY\)](#), Jenny List(DESY), Junping Tian(U of Tokyo)

Follow up 2511.08359



Main physical processes at Linear Collider Facility (LCF)

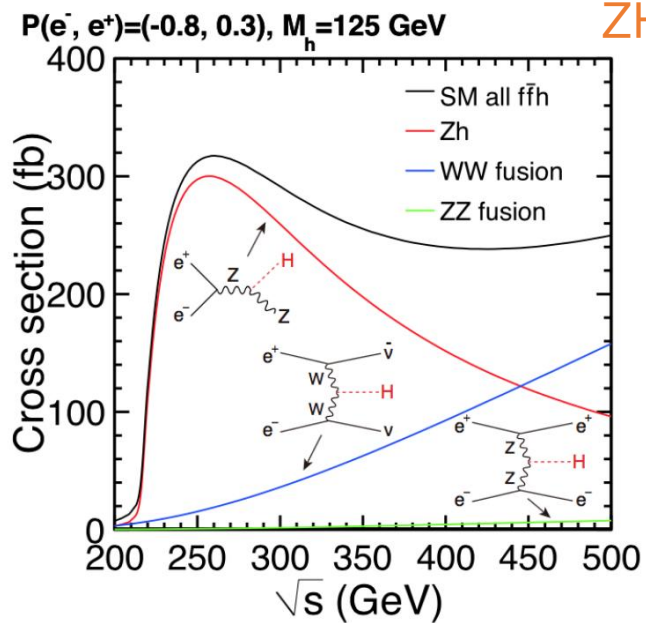
LC vision. Eur. Phys. J. Spec. Top. (2026).

FCC-ee

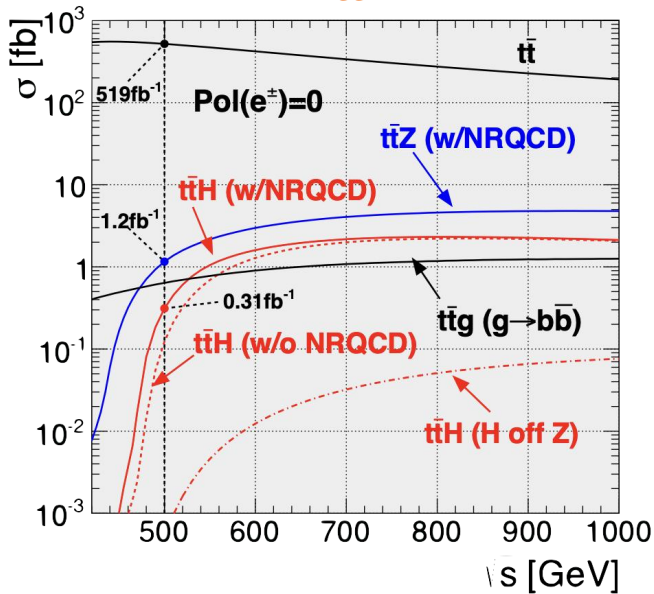
	91 GeV	250 GeV	350 GeV	550 GeV	1-3 TeV
$\int \mathcal{L} \text{ (ab}^{-1}\text{)}$	0.1	3	0.4	8	8
beam polarisation (e^-, e^+ ; %)	80/30	80/30	80/30	80/60	80/20
($--, -+, +-, ++$) (%)	(10,40,40,10)	(5,45,45,5)	(5,68,22,5)	(10,40,40,10)	(10,40,40,10)

ZH

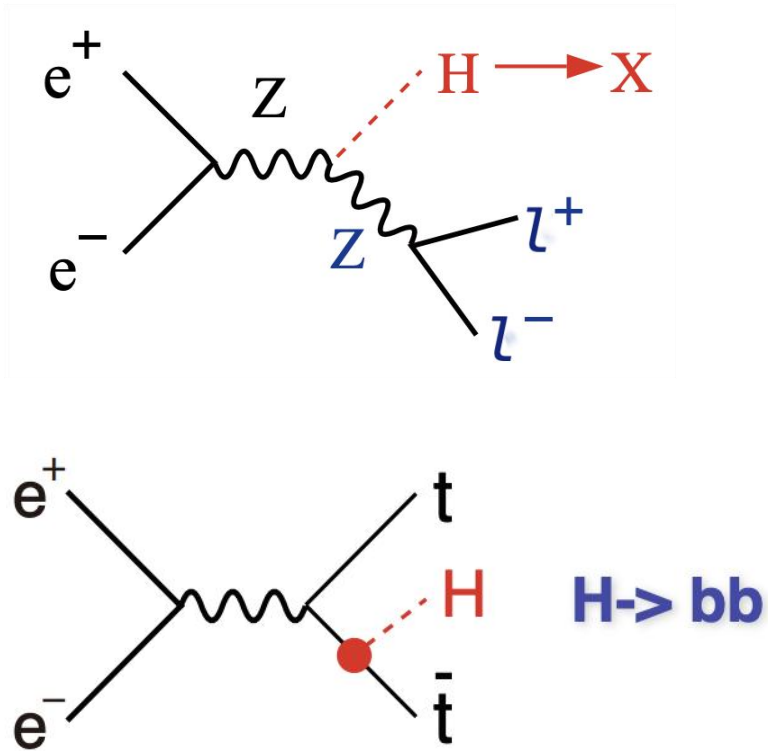
ttH



H. Baer et al., eds., 1306.6352.

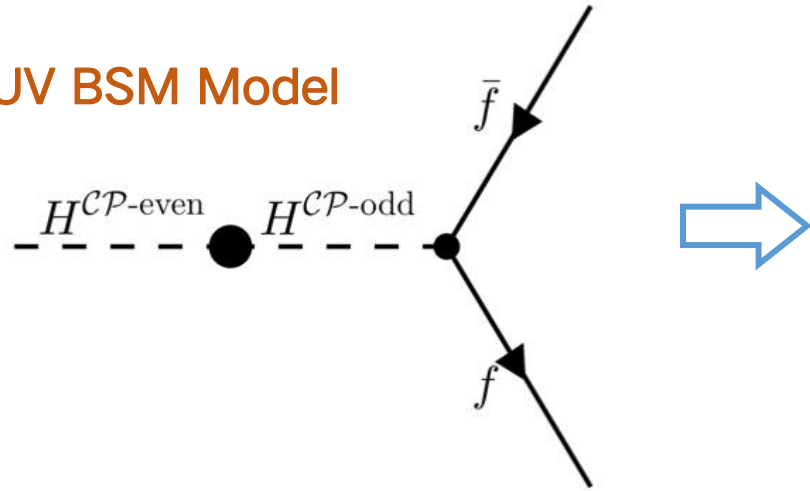


Yonamine, et al., PRD84, 014033



Theoretical CPV model : SMEFT

UV BSM Model



$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_{d>4} \sum_i^{N_d} \frac{C_i O_i^{(d)}}{\Lambda^{d-4}}$$

$d=6$
 CPV
 1TeV

ZH ttH

$$O_{\tilde{G}}, O_{\tilde{W}}, O_{\varphi\tilde{G}}, O_{\varphi\tilde{W}}, O_{\varphi\tilde{B}}, O_{\varphi\tilde{W}B}, \hat{O}_{t\varphi}, \hat{O}_{tG}, \hat{O}_{tW}$$

ZH

$$\begin{aligned} C_{H\tilde{Z}\tilde{Z}} &= c_w^2 C_{H\tilde{W}} + s_w (c_w C_{H\tilde{W}B} + s_w C_{H\tilde{B}}) \\ C_{H\tilde{A}\tilde{A}} &= c_w^2 C_{H\tilde{B}} + s_w (s_w C_{H\tilde{W}} - c_w C_{H\tilde{W}B}) \\ C_{H\tilde{A}\tilde{Z}} &= 2s_w c_w (C_{H\tilde{W}} - C_{H\tilde{B}}) + (s_w^2 - c_w^2) C_{H\tilde{W}B} \end{aligned}$$

ttH

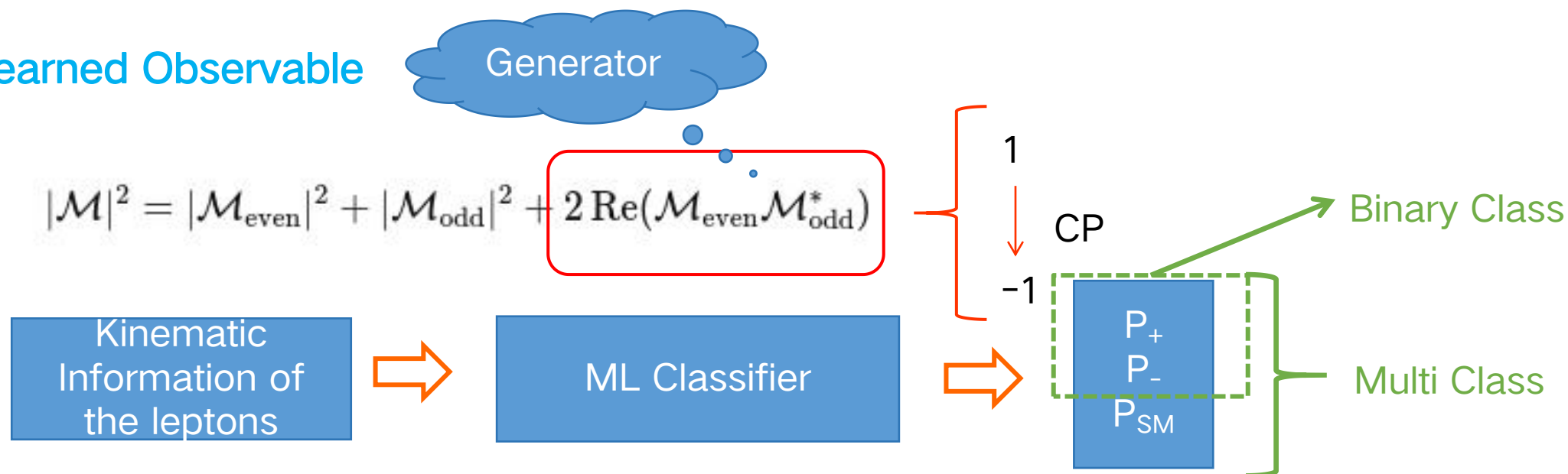
$$\begin{aligned} \mathcal{L}_{h\bar{t}t} &= -\frac{m_t}{v} \bar{t} (\kappa \cos \alpha + i\gamma_5 \kappa \sin \alpha) t h, \\ \kappa \cos \alpha &= 1 - \frac{v^3}{\sqrt{2} m_t} \frac{C_{t\varphi}}{\Lambda^2}, \quad \kappa \sin \alpha = -\frac{v^3}{\sqrt{2} m_t} \frac{C_{t\varphi}^I}{\Lambda^2}. \end{aligned}$$

CP observables: Traditional vs Machine Learning

Angular Observable(e.g. in ZH)

$$\mathcal{M}_{\text{odd}} \sim \Delta\phi_{\ell\ell}, \quad \Delta\phi_{\ell\ell} \equiv \phi(\ell_1) - \phi(\ell_2), \quad \eta(\ell_1) > \eta(\ell_2), \quad Z \rightarrow \ell^+ \ell^-, \quad \ell = e, \mu$$

Optimal ML-learned Observable



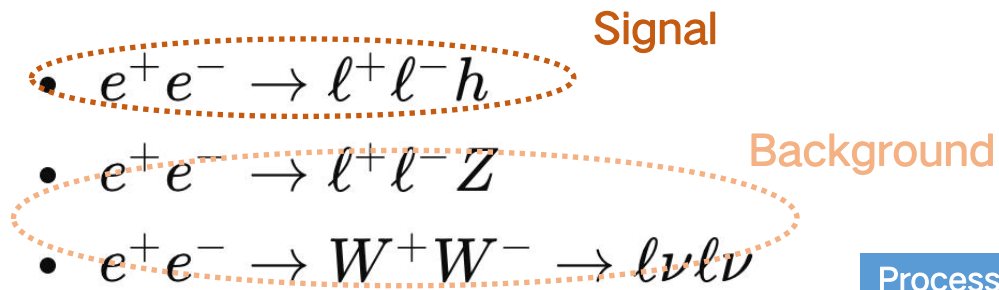
$$O_{\text{ML}} = P_+ - P_- \quad -1 \leq O_{\text{ML}} \leq 1$$

Binary Class: $P_+ + P_- = 1$ Multi Class: $P_+ + P_- + P_{\text{SM}} = 1$

ZH Analysis Setup

MC Simulation

- Generator: MadGraph5+Pythia8 + SMEFTSim3.0



- Detector Simulation: Delphes v3
 - ILCgen card for LCF
 - IDEA card for FCC-ee

Selection

$|M_{\ell\ell} - M_Z| < 15 \text{ GeV}$
 $|M_{\text{recoil}} - M_H| < 5 \text{ GeV}$

$p_T^{\ell\ell} > 10 \text{ GeV}$
 $\arccos(\vec{n}_{\ell_1} \cdot \vec{n}_{\ell_2}) > 100^\circ$

$|p_z^{\ell\ell}| < 70 \text{ GeV}$
 $\arccos((\vec{n}_{\ell_1} \times \vec{n}_{\ell_2}) \cdot \hat{k}) > 10^\circ$

Process	lepton pair	cross section/fb	Efficiency
ZH	mu	15.0518	86.3%
	e		84.4%
ZZ	mu	225.065	0.47%
	e		0.06%
WW	mu	707.35	1.00%
	e		0.96%

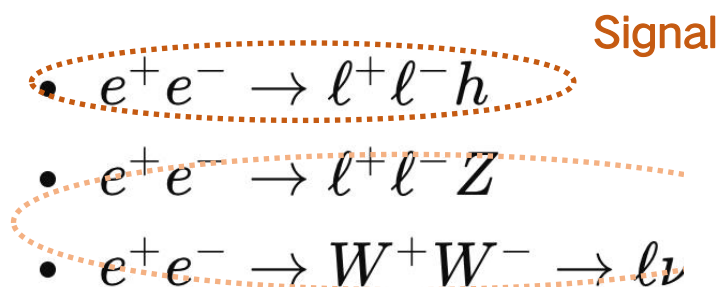
$C_{\text{EFT}} / \Lambda^2=1$	lepton pair	Efficiency
$C_{H\tilde{W}}$	mu	84.3%
	e	70.4%
$C_{H\tilde{W}B}$	mu	79.3%
	e	62.5%
$C_{H\tilde{B}}$	mu	78.8%
	e	60.5%



ZH Analysis Setup

MC Simulation

- Generator: MadGraph5+Pythia8 + SMEFTSim3.0



- Detector Simulation: Delphes v3
 - ILCgen card for LCF
 - IDEA card for FCC-ee

Selection

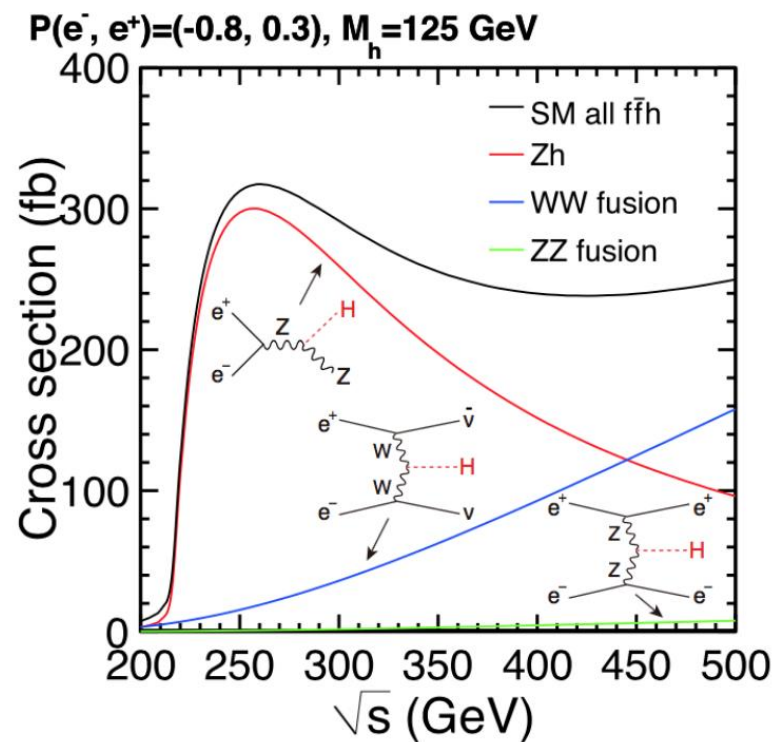
$$|M_{\ell\ell} - M_Z| < 15 \text{ GeV} \quad |M_{\text{recoil}} - M_H| < 5 \text{ GeV}$$

$$p_T^{\ell\ell} > 10 \text{ GeV}$$

$$\arccos(\vec{n}_{\ell_1} \cdot \vec{n}_{\ell_2}) > 100^\circ$$

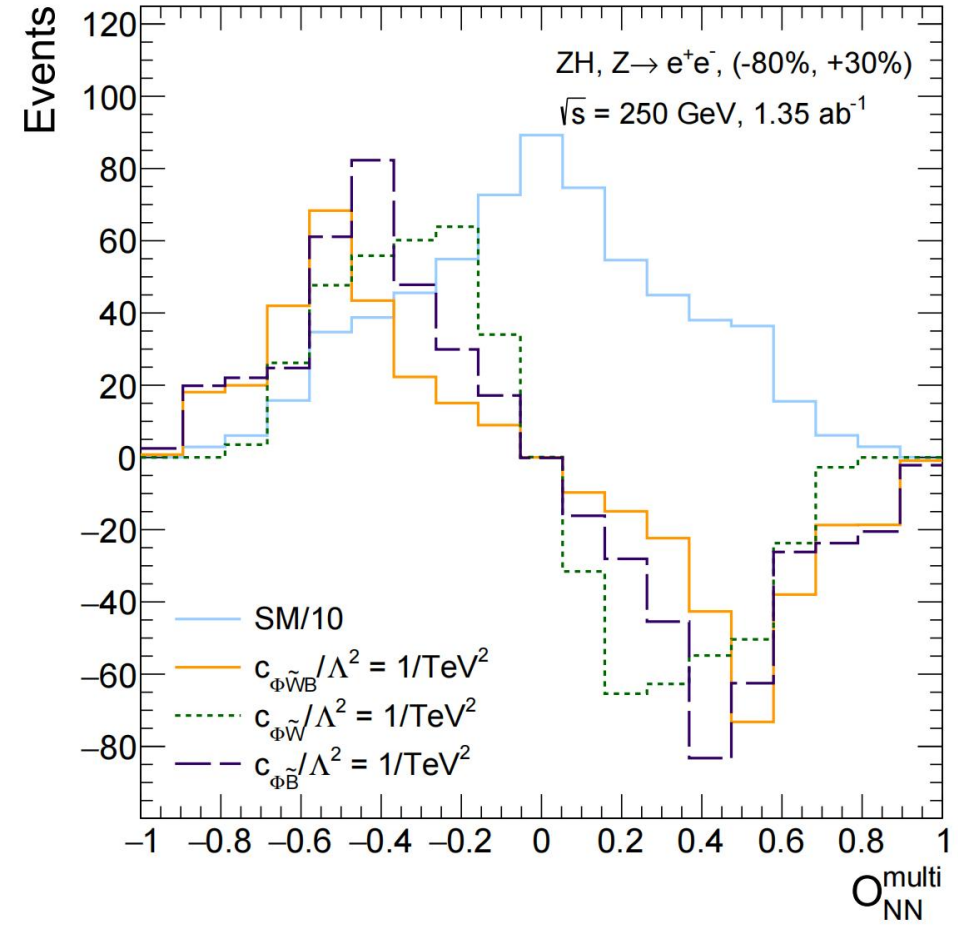
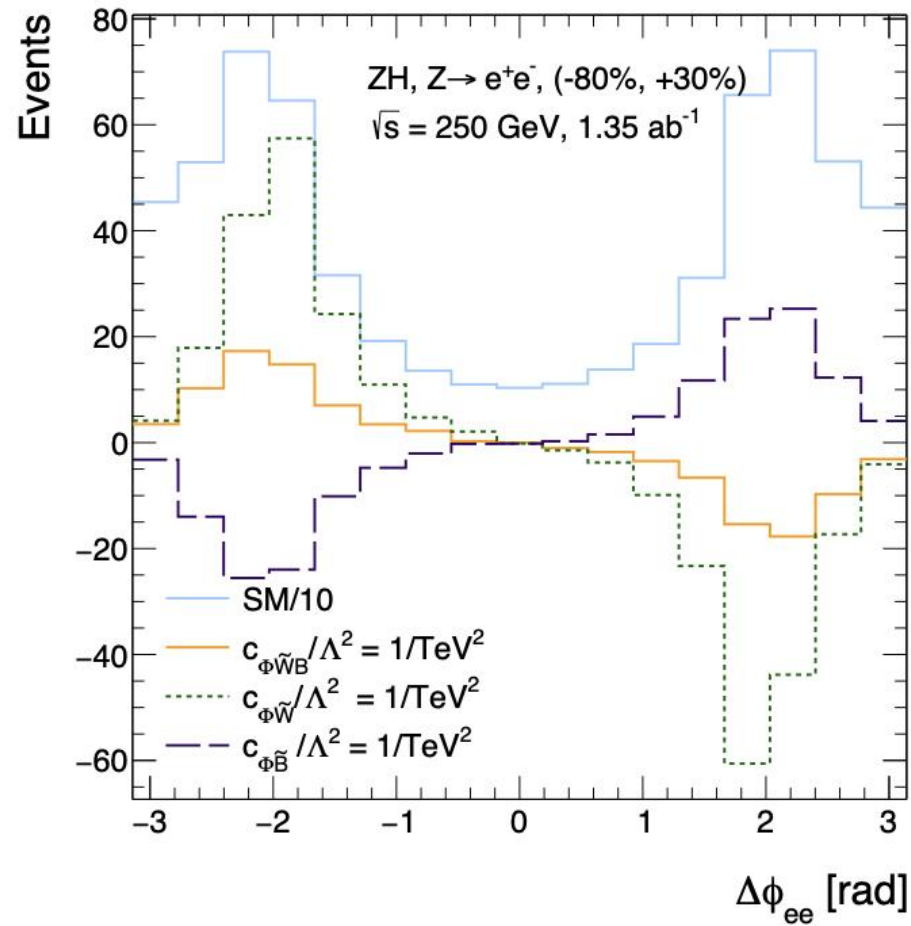
$$|M_{\ell\ell}| > 70 \text{ GeV}$$

$$\arccos((\vec{n}_{\ell_1} \times \vec{n}_{\ell_2}) \cdot \hat{k}) > 10^\circ$$

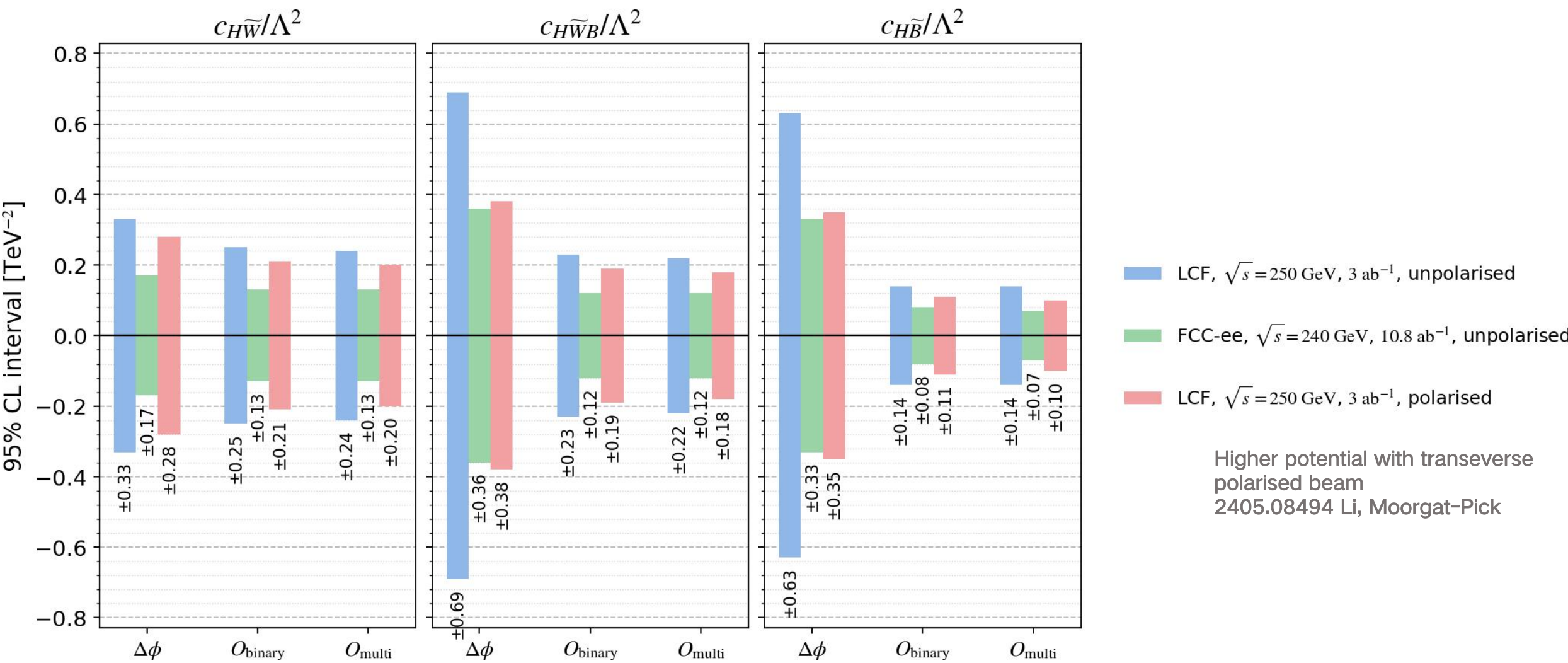


$C_{\text{EFT}} / \Lambda^2=1$	lepton pair	Efficiency
$C_{H\tilde{W}}$	mu	84.3%
	e	70.4%
$C_{H\tilde{W}B}$	mu	79.3%
	e	62.5%
$C_{H\tilde{B}}$	mu	78.8%
	e	60.5%

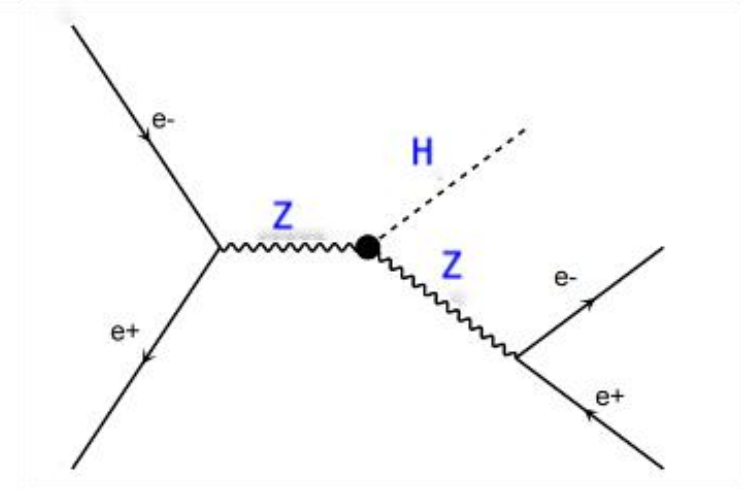
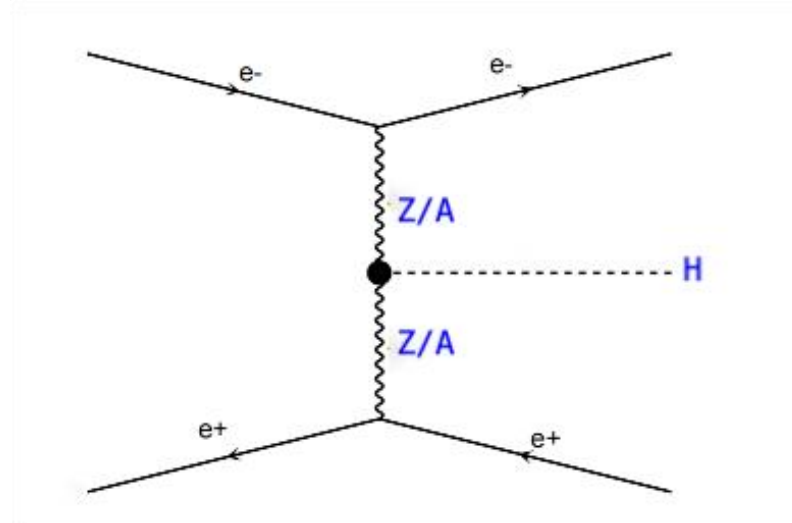
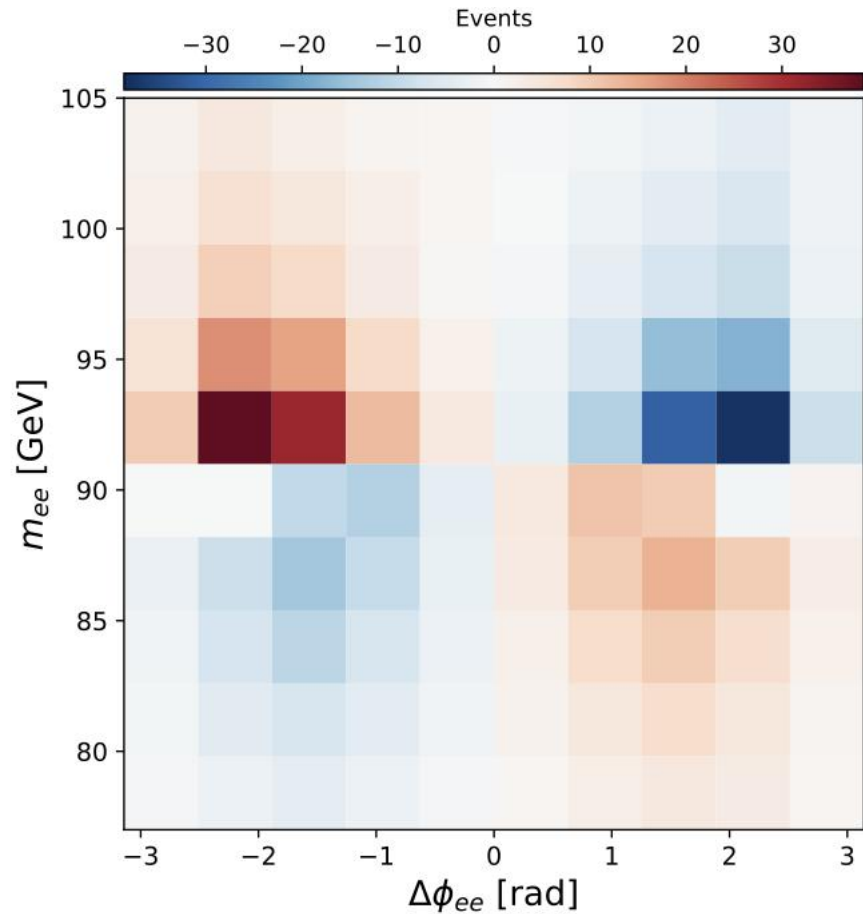
ZH: Comparison of Angular Observable and ML-learned Observable



ZH: Projected 95% CL intervals for CP-odd SMEFT Wilson coefficients



ZH: Decode the ML Black Box: The interference structure



ttH at 550GeV: Ongoing

Observable Development

$$\sin \phi = \frac{\vec{p}_2(\vec{q}_a \times \vec{p}_1)}{|\vec{p}_2||\vec{q}_a \times \vec{p}_1|}$$

top, anti-top

Godbole, 1103.5404

ZH tth interference

MC Simulation

- Generator: Physsim + SMEFTSim3.0
 - NRQCD bound-state effect included
 - Decayed final state with resonant top propagators (6 fermions + H)
- Detector Simulation: SGV

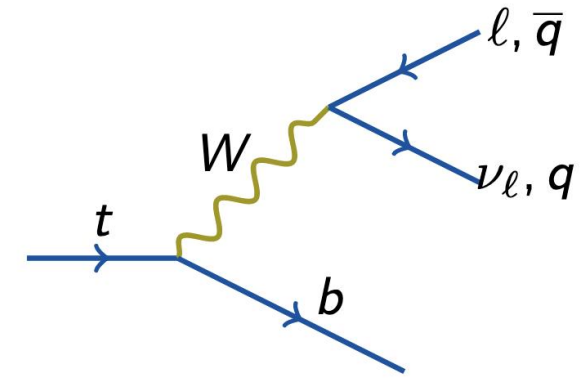
particle/anti-particle
decay from W

Ma, 1807.07127

tt spin correlation

Potential Improvement

ML Input Features

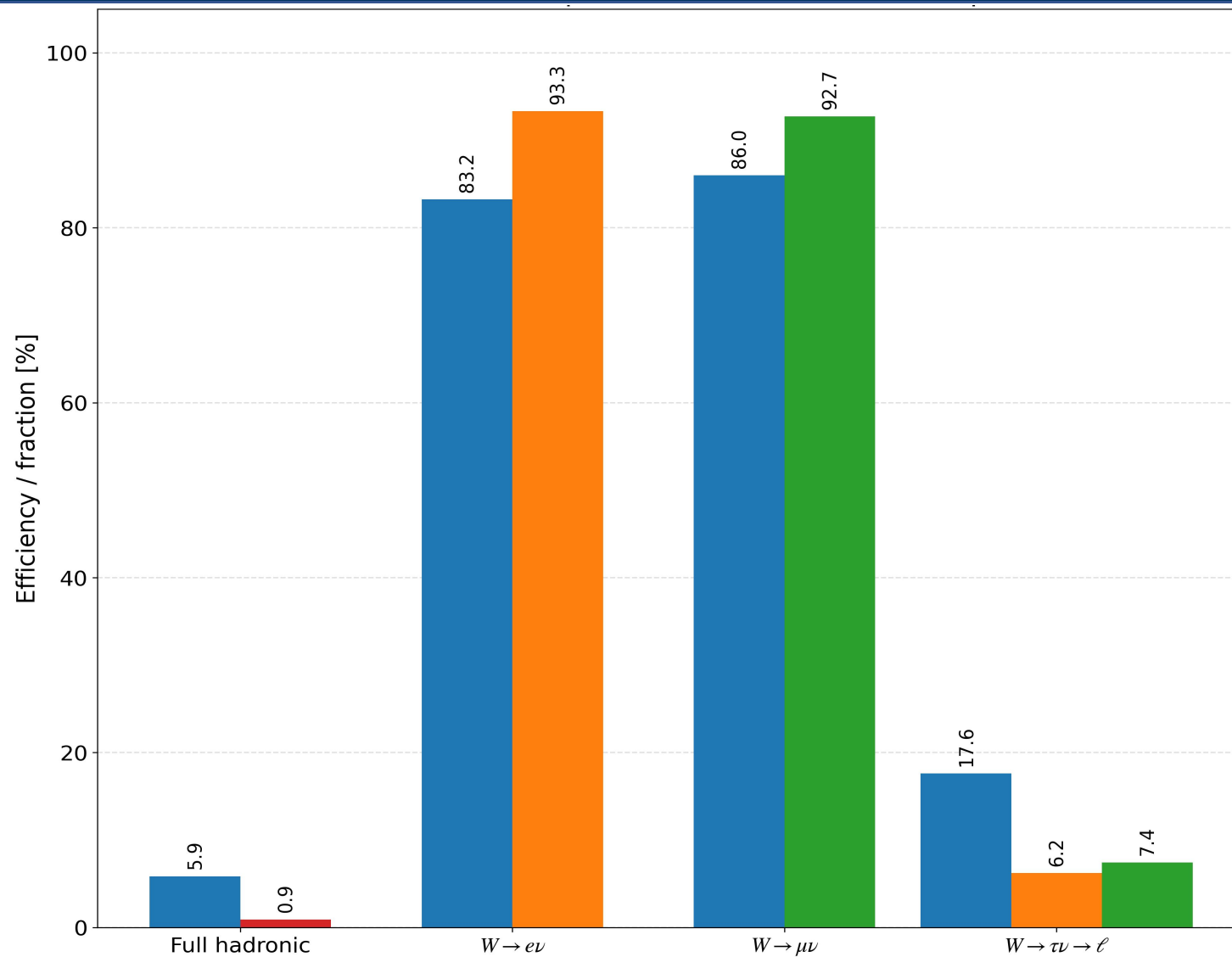


- TMVA Isolated Lepton Tagger, 2018
 - 46% Semi-leptonic events
 - 9% Di-leptonic events
- Particle Transformer(ParT) for Jet Tagging, 2024
 - 45% Full-hadronic events

ttH at 550GeV: Isolated Lepton Tagger Implementation in ttH

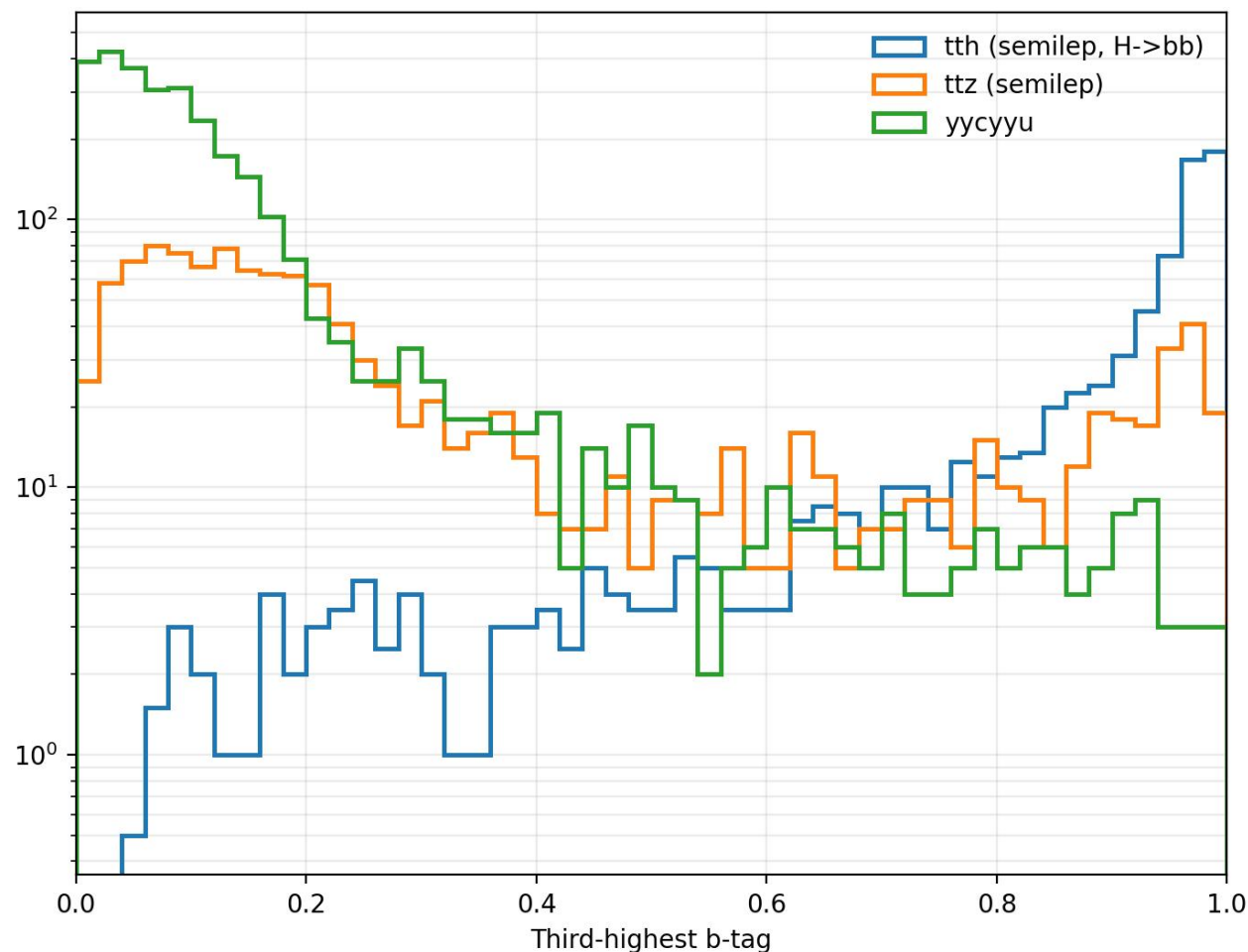
H→bb, semilepton

- Cut-based: 1 Isolep
- TMVA: 1 ISOElectron
- TMVA: 1 ISOMuon
- TMVA: 1 ISOElectron or ISOMuon

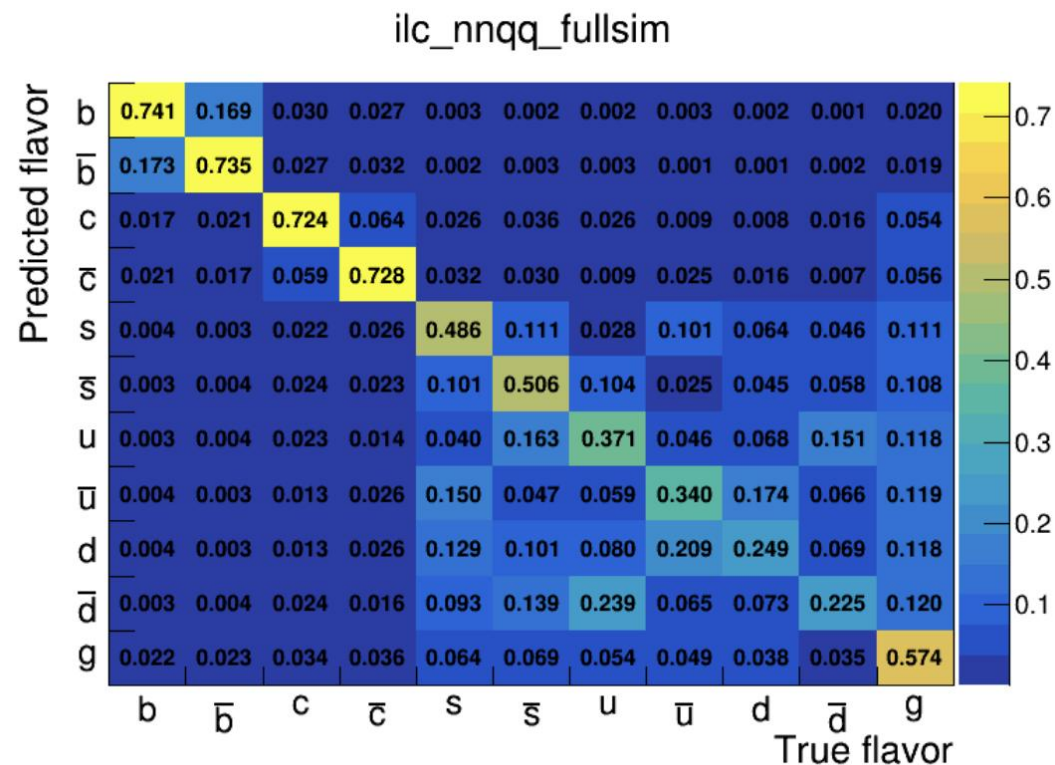


ttH at 550GeV: ParT Flavor Tag Applied in Linear Collider

Improved b-jet selection



The ability of identifying jet charge



Taikan Suehara et al., LCWS2025

Summary

Established in ZH

- CP-odd observables directly probe the interference term
- ML-learned observables outperform the azimuthal angle
- LCF polarisation partially compensates lower luminosity

Ongoing in ttH

- Isolated-lepton reconstruction baseline validated
- ParT flavor information will be used
- Next step: Evaluate the performance of ML-based CP observable in ttH analysis

Thank you for attention!



University
of Glasgow



Backup



University
of Glasgow



Backup

Limit Setting

The profile likelihood ratio

$$\lambda(c) = -2 \log \frac{L(c)}{L_{\max}} \quad L(c) = \prod_{i=1}^n \frac{e^{-\lambda_i}}{n_i!} \lambda_i^{n_i}$$

where n_i is the number of observed events and λ_i is the prediction under a given model with parameter c .

The theoretical number $\lambda_i(c)$ will be evaluated by

$$\lambda_i(c) = \lambda_i^{\text{SM}} + c \cdot \lambda_i^{\text{int}}(c=1) + c^2 \cdot \lambda_i^{\text{EFT}}(c=1)$$

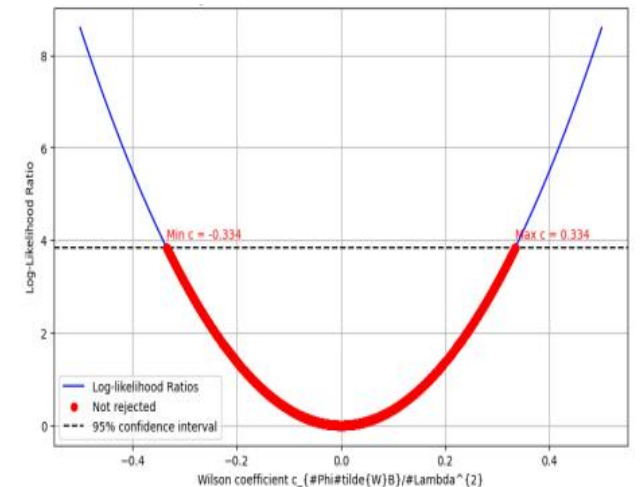
Combine the channels

$$L_{\text{combined}}(c) = L_{\mu\mu}(c) L_{ee}(c)$$

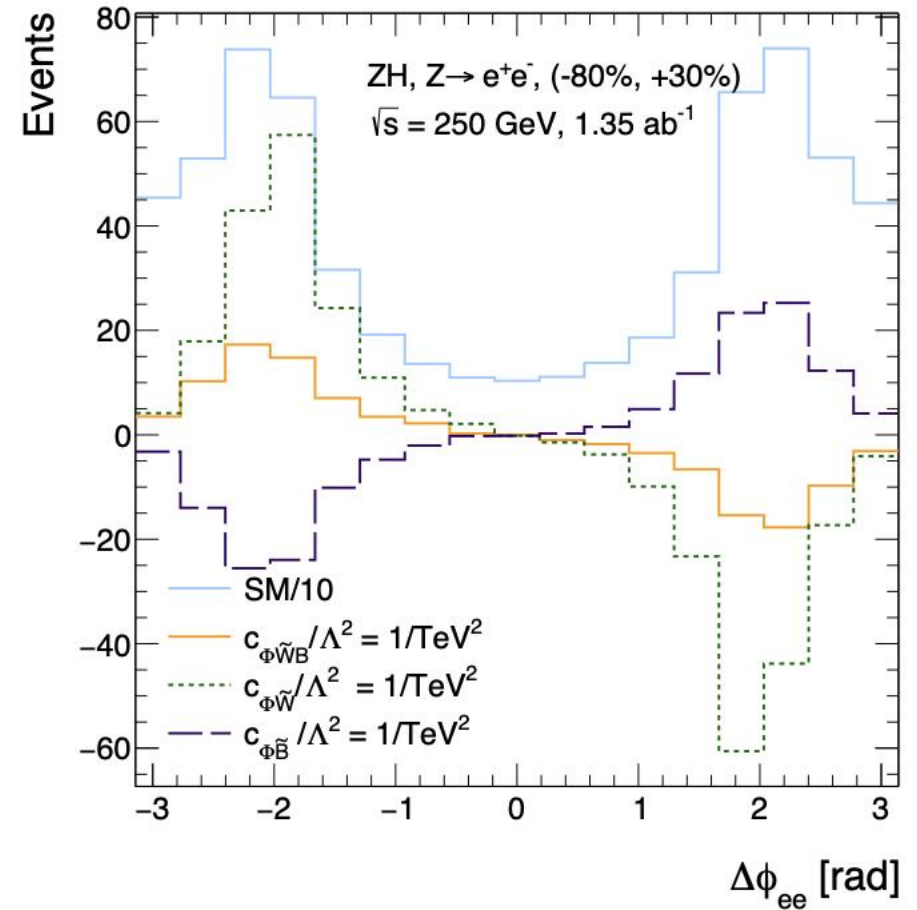
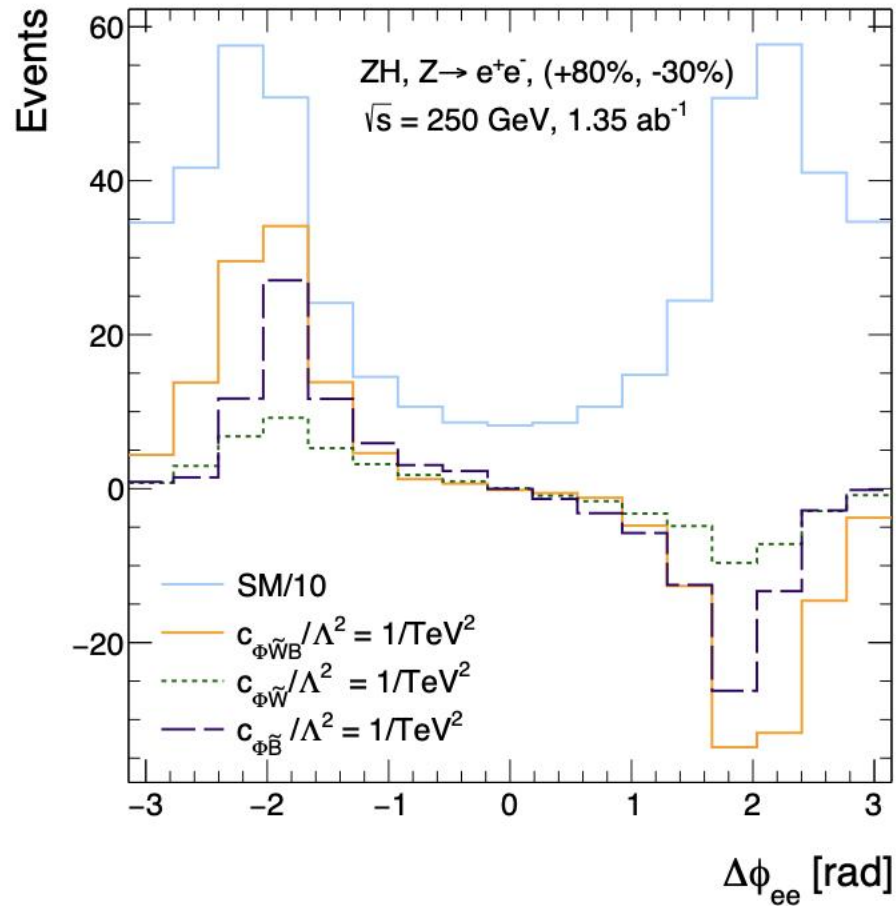
95% Confidence Interval

Following the Wilks' theorem with 1 d.o.f.,

$$\lambda_{\text{all}}(c) = -2 \sum_{i=1}^4 \log \left(\frac{\mathcal{L}_{i,\mu\mu}(c) L_{i,ee}(c)}{\mathcal{L}_{i,\mu\mu,\max} L_{i,ee,\max}} \right) < 3.84$$



Backup: Angular observable with Polarised Beam



Backup: Angular observable with Polarised Beam

